Evidence for a Normal Geomagnetic Field Polarity Event at 263± 5 m.y. BP within the Late Palaeozoic Reversed Interval

K. M. CREER & J. G. MITCHELL
Department of Geophysics and Planetary Physics, University of Newcastle upon Tyne

D. A. VALENCIO
Department of Geological Sciences, University of Buenos Aires

Basalts, radiometrically dated at 263±5 m.y., have stable remanent magnetization with normal polarity. This demonstrates the existence of a previously unrecognized event within the Late Palaeozoic Interval of reversed geomagnetic polarity.

THE Late Palaeozoic Interval of reversed geomagnetic polarity (the Kiaman) lasted for about 60 m.y. during the Upper Carboniferous and Permian periods. As originally recognized in Australia by Irving and Parry[1] the base of the Kiaman division rests stratigraphically just above the Paterson toscanite (age 298 m.y.) which has normal remanent magnetization (NRM). In Europe this corresponds to the Westphalian–Stephanian boundary and in the United States to the top of the Desmoinesian. In the USSR, however, Khramov[2] placed the base at 305 to 310 m.y. BP, below a normal event recognized in the Moscovian of the Donbas section to which the Australian Paterson reversal corresponds. The end of the late Palaeozoic Reversed Interval is generally recognized as being marked by the onset of frequent reversals in the topmost Permian at about 230 to 235 m.y. BP.

Normal polarity within the interval between 290 and 235 m.y. BP has only been observed rarely and even then the magnetic stability of the rocks has not been well established and the age has not been well determined. Both in the USSR and the USA there is some evidence of a normal event, placed at 280 m.y. BP by McElhinny[3] who named it the "Oak Creek" event. Burek[5] named this same event the "Graham" event. (The reason for the choice of these two names for this event was that Graham[4] had discovered normal magnetization in the Permian Supai Formation at Oak Creek Canyon, Arizona.) The only other study suggesting the presence of normal polarity between 290 and 235 m.y. BP is that of Middle Permian sediments from Italy[6] but these were not subjected to magnetic cleaning.

Porphyritic Series

The Porphyritic Series has been described by Groeber[7]. It consists of basalts, andesites and rhyolites and is exposed in the northern part of Mendoza Province, Argentina. Dessanti[8] and Dessanti and Caminos[9] have studied these igneous units more recently, and at Cuesta de los Terneros, about 20 km SSW of San Rafael City, they found that the lavas of this series overlie the Lower Permian Cochico Formation unconformably. In this region the Porphyritic Series is overlain by the Puesto Viejo Formation for which Diaz Gonzales[10] suggested a Middle Triassic age, and later Bonaparte[11] suggested, more precisely, a lower Anisian age (about 210 m.y.). Thus according to the available geological information, the age of the lavas of the Porphyritic Series in this area from which the palaeomagnetic collection (section 3) was made must be between Lower Permian (about 270 m.y.) and Lower Anisian (about 210 m.y.).

Valencio[12] describes palaeomagnetic studies of samples of the Porphyritic Series from four areas situated between 20 km and 80 km to the south of San Rafael City. At one of these areas, Rio Seco Los Leones, the NRM was unstable, but at the other three the NRM exhibited stable characteristics.

At the two collecting areas near the Valle Grande Dam on the Rio Atuel, close to the area studied geologically[8,9], the rock type was basalt and the NRM had blocking temperatures of about 580° C. Exclusively normal polarities were encountered in fifteen samples from area 3 and ten samples from area 4. The stability of the RM was tested by thermal demagnetization and the means of the cleaned directions of samples from these areas are shown in Table 1.

At area 1, six normally magnetized samples and ten reversely magnetized samples of rhyolite and andesite were measured and the means of the cleaned RM directions from this area are also shown in Table 1.

Other Formations

The positions of the south virtual geomagnetic poles (VGPs) obtained for the Porphyritic Series are compared with the Late Palaeozoic–Early Mesozoic polar wandering curve for South America in Fig. 1. Also plotted in Fig. 1 are the south VGPs obtained from exclusively reversely magnetized Late Palaeozoic formations exposed nearby. The poles[13] for the lower levels of the Paganzo formation at Los Colorados (La Rioja Province) and for the same formation at Huaco Gorge are represented by the black triangles labelled respectively *LC*1 and *H*. These VGPs are included in the population of six formation VGPs used to derive the palaeomagnetic pole for South

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America in the Upper Carboniferous[14], which is represented by the large black dot labelled C_u in Fig. 1. The upper levels of the Paganzo formation at Los Colorados yield the VGP labelled *LC2* which is one of the population of three formation VGPs accepted for calculating[14] the South American Permian palaeomagnetic pole (large dot labelled *P*).

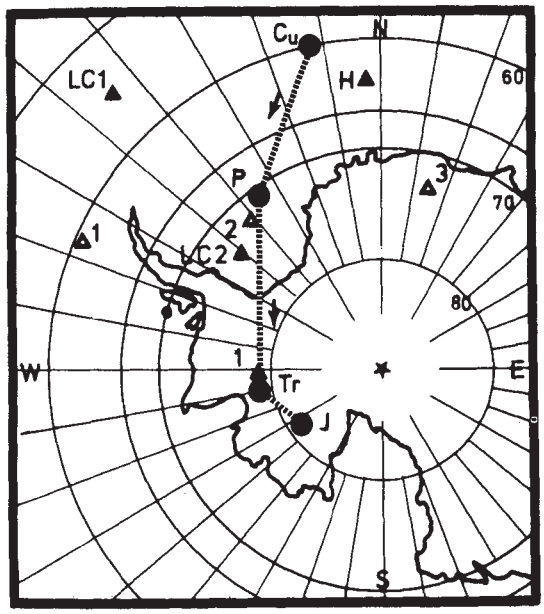


Fig. 1 – – –, Part of the polar wander curve for South America connecting the palaeomagnetic south poles (●) for the Late Palaeozoic Interval of reversed geomagnetic polarity (C_u, Upper Carboniferous; *P*, Permian) to the Triassic and Jurassic palaeomagnetic poles, *Tr* (including some data from the topmost Permian) and *J* respectively[3]. ▲, South VGPs for formations with reversed RM; △, south VGPs for formations with normal RM; *LC*1 and *LC*2, lower and upper levels of Paganzo formation at Los Colorados; *H*, Paganzo formation at Huaco Gorge; 1, 2 and 3, Porphyritic Series.

One of the reasons for plotting these formation VGPs is to illustrate some typical deviations from the respective palaeomagnetic pole which is the mean of the population of VGPs of formations of a restricted age range (which need not necessarily correspond to a geological period). This provides some criteria for deciding whether the magnetic age of the VGPs derived from the Porphyritic Series is Upper Carboniferous, Permian or Triassic.

The palaeomagnetic pole labelled *Tr* is the mean of three VGPs from formations of Triassic (possibly Late Permian) geological age.

The VGPs for the exclusively normally magnetized basalts collected from near the Valle Grande Dam are represented by the open triangles labelled 3 and 4 in Fig. 1, and their position is indicative of magnetization in the Permian field rather than in the Triassic field as originally supposed[12] (see later).

The normal RM of the andesites and rhyolites at Valencio's area 1 at Rio Seco Los Toldos also seems to be characteristically Early Permian rather than Permo-Triassic (the VGP is represented by the open triangle labelled 1 in Fig. 1). But the reversed RM encountered in samples collected from this area, for which the VGP is represented by the black triangle labelled 1, is clearly characteristic of the Permo-Triassic. This is still consistent with the younger geological age limit of Lower Anisian[11].

The conclusion we wish to draw attention to is, however, that the normal RM at the Valle Grande Dam sites is characteristic of the South American Permian palaeomagnetic field deduced from the palaeomagnetism of rock formations exhibiting exclusively reversed RM.

Table 2 K-Ar Analyses of Normally Magnetized Samples from near Valle Grande Dam

Sample number*	K_2O %	Radiogenic argon $mm^3\ g^{-1}$	Age m.y.
1248 (226)	2.63 ± 0.04	$(2.44 \pm 0.05)\ 10^{-2}$	262 ± 6
1252 (225)	2.43 ± 0.05	$(2.28 \pm 0.04)\ 10^{-2}$	264 ± 6
$\lambda_B = 4.72 \times 10^{-10}\ yr^{-1}$ $\lambda_C = 0.584 \times 10^{-10}\ yr^{-1}$		$^{40}K/K = 1.19 \times 10^{-2}$ atom %	

* Sample No.: the first number is that allotted by Valencio[12] in his palaeomagnetic study. The second number, in parentheses, is the reference number allotted by the K–Ar geochronological laboratory at the University of Newcastle.

Radiometric Age

The potassium–argon ages of samples from Valle Grande Dam (area 3) were determined by conventional isotope dilution using an omegatron mass spectrometer[15] and flame photometric analysis for potassium. Mean potassium content is calculated from seven analyses and argon content from two analyses. The analytical data are presented in Table 2.

Agreement between the ages of the independent samples can be taken as evidence supporting the validity of the isotopic age. Effects such as argon loss and "excess" argon are more likely to produce discordant ages from samples of the same geological age. The mean potassium–argon age is therefore 263 ± 5 m.y.

The geological and palaeomagnetic evidence then available led Valencio[12] to suggest that the volcanic rocks of the Porphyritic Series were post-Kiaman because of their mixed normal and reversed polarity. The radiometric ages of samples 1248 and 1252 described here, together with the stratigraphic control described above that the Series be later than Lower Permian, indicate, however, that it was formed during the Kiaman interval. Moreover the normal polarity event so defined is distinguishable both by stratigraphic and isotopic evidence from the Permo-Carboniferous normal polarity event at 280 m.y. (the so called "Oak Creek" or "Graham" event).

More Late Palaeozoic Formations

The Peruvian Mitu formation consisting of red sediments with intercalated igneous bodies has been studied palaeomagnetically. Although the NRM directions do not fit the

Table 1 Palaeomagnetic Directions of Porphyritic Series, Mendoza Province

Area	Rock type	*N*	Area-mean directions of cleaned RM *D*	*I*	*R*	*K*	α	Polarity
(1) Rio Seco Los Toldos	Andesites and rhyolites	6	358°	−76°	5.96	145	5.5°	N
		10	186°	+63°	9.81	48	7.1°	R
(3) Near Valle Grande Dam	Basalt	15	339°	−55°	14.6	36	6.5°	N
(4) Near Valle Grande Dam	Basalt	10	348°	−67°	9.9	88	5.2°	N

N, No. of samples; *R*, resultant vector; *K*, precision; α, radius of 95% circle of confidence; *D*, *I*, declination and inclination of cleaned RM of area mean; polarity: N, normal; R, reversed. The collecting areas are near 33° S 68° W.

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polar wander curve for South America, possibly because of tectonic affects of the Andean orogeny, there is clear evidence of mixed polarity. The NRM has been thermally cleaned[16]. Stratigraphically, the Mitu formation is placed in the Permian and, furthermore, recent K–Ar age determinations on associated igneous bodies by Rocha-Campos and Amaral[17] yield ages of 260 m.y.

Thus the normal magnetization recorded in the Mitu formation may also correspond to the normal event recorded by the basalts of the Porphyritic Series from Mendoza Province, Argentina.

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Accurate Radio and Optical Positions of 3C273B

C. HAZARD & J. SUTTON
Institute of Theoretical Astronomy, University of Cambridge

A. N. ARGUE & CECILIA M. KENWORTHY
The Observatories, University of Cambridge

L. V. MORRISON & C. A. MURRAY
Royal Greenwich Observatory, Herstmonceux Castle, Sussex

New measurements have confirmed that 3C273B and the associated quasi-stellar object are coincident.

THE small angular size component (3C273B) of the radio source 3C273 is generally considered to coincide with the associated quasi-stellar object (QSO), and the most accurate published optical and radio positions[1] are consistent with this interpretation. Nevertheless, they do differ by 0.7 arc s, a discrepancy which is just outside the combined estimated errors, and the possibility of a small but significant displacement between the QSO and 3C273B cannot be excluded. In an attempt to reduce this uncertainty we have therefore carried out a new measurement of the position of 3C273B based on occultation observations using the CSIRO 210-foot telescope at Parkes, Australia, and the 1,000-foot telescope at Arecibo, Puerto Rico. We had not previously carried out this analysis because it was not possible to realize the full positional accuracy of which these high signal/noise observations were capable because of uncertainties in the Improved Lunar Ephemeris and in the difference (ΔT) between Ephemeris (ET) and Universal Time (UT). Recently, corrections to the Improved Lunar Ephemeris[2] have been proposed and, with accurate values for ΔT over the period 1963 to 1966 also now available, it seemed possible to reduce the error in the radio position to ≤ 0.2 arc s. As it turned out, our new position measurement only increased the discrepancy between the optical and radio positions. On examining the optical position, however, we found that the usually quoted value was based on observations made by Jefferys[1] for the purpose of proper motion studies and that no accurate optical position had been measured. We therefore carried out new position measurements of the QSO both at the Cambridge Observatories and at the Royal Greenwich Observatory (RGO) to permit an accurate comparison with the new radio position.

Radio Position

The method of observation and an account of the structural information obtained from an analysis of all the occultation curves have been given elsewhere[3,4]. For the present investigation we selected only those occultation curves for which the signal/noise ratio and the separation of the A and B components permitted a timing accuracy of better than 1 s, corresponding to an error in the location of the Moon's mean limb of ≤ 0.3 arc s. The relevant data relating to these selected curves are listed in Table 1 together with the estimated times of occultation, all of which have been corrected for delays introduced by the output time constants of the receiving equipment. In the case of the grazing occultation the quoted occultation time corresponds to the time of closest approach. For the other occultations the occultation times were determined using the restoration technique described by Scheuer[5], which gives the brightness distribution across the source as seen by a fan beam of arbitrary width. The widths of the restoring beams used ranged from a minimum of 0.18 arc s for the occultation of June 7, 1965, to a maximum of 3.5 arc s for the occultation of September 7, 1964, where the occultation occurred during the day and close to the Sun, and the observa-

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